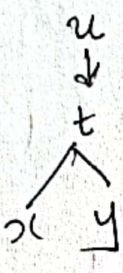
* Differentiation of a function of a function



. If u is a function of t, where t is a function of x and y then

$$\frac{\partial u}{\partial x} = \frac{du}{dt} \cdot \frac{\partial t}{\partial x} \quad \text{and}$$

$$\frac{\partial u}{\partial y} = \frac{du}{dt} \frac{\partial t}{\partial y}.$$

If $x = r\cos\theta$, $y = r\sin\theta$ then $r^2 = x^2+y^2$ & $\tan\theta = \frac{y}{x}$

$\therefore \left(\frac{\partial x}{\partial r}\right)_\theta = \cos\theta, \quad \left(\frac{\partial x}{\partial \theta}\right)_r = -r\sin\theta$

$$\left(\frac{\partial x}{\partial r}\right)_y = \frac{\partial}{\partial r}\sqrt{r^2-y^2} = \frac{1}{2}(r^2-y^2)^{-\frac{1}{2}}(2r) = \frac{r}{\sqrt{r^2-y^2}} = \frac{r}{\sqrt{r^2(1-\sin^2\theta)}}$$

$\therefore \left(\frac{\partial x}{\partial r}\right)_y = \sec\theta, \quad \left(\frac{\partial y}{\partial r}\right)_x = \frac{\partial}{\partial r}\sqrt{r^2-x^2} = \operatorname{cosec}\theta$

$$\left(\frac{\partial y}{\partial x}\right)_\theta = \frac{\partial}{\partial x}(x\tan\theta) = \tan\theta$$

$$\left(\frac{\partial x}{\partial \theta}\right)_y = \frac{\partial}{\partial \theta}\left(\frac{y}{\tan\theta}\right) = \frac{\partial}{\partial \theta}(y\cot\theta) = -y\operatorname{cosec}^2\theta$$

$$= -\frac{r\sin\theta}{\sin^2\theta}$$

$$= -\frac{r}{\sin\theta}$$

$\therefore \left(\frac{\partial x}{\partial r}\right)_\theta \neq \left(\frac{\partial x}{\partial r}\right)_y$ and $\left(\frac{\partial x}{\partial \theta}\right)_r \neq \left(\frac{\partial x}{\partial \theta}\right)_y$.

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Variables To be Treated As Constants:

In some problems, it is difficult to identify which variable is to be treated as constant.
In such cases, the variable to be treated as constant is written as the suffix of the bracket.

$\left(\frac{\partial r}{\partial x}\right)_y$ - First r is to be expressed as a function of x & y and then differentiated w.r. to x keeping y constant

Ily, $\left(\frac{\partial x}{\partial r}\right)_\theta$ - First express x as a function of r & θ and then differentiate w.r. to r keeping θ constant

Classroom problems:

If $x^2 = au + bv$, $y^2 = au - bv$, Prove that $\left(\frac{\partial u}{\partial x}\right)_y \left(\frac{\partial x}{\partial u}\right)_v = \frac{1}{2} = \left(\frac{\partial v}{\partial y}\right)_x \left(\frac{\partial y}{\partial v}\right)_u$, where a and b are constants.

Here $x^2 = au + bv$ — (1)

$y^2 = au - bv$ — (2)

For $\left(\frac{\partial u}{\partial x}\right)_y$ - express first u as $f(x, y)$ then differentiate w.r. to x keeping y constant.

eqn (1) + (2),

$x^2 + y^2 = 2au$

$\therefore u = \frac{x^2 + y^2}{2a}$

$\therefore \left(\frac{\partial u}{\partial x}\right)_y = \frac{1}{2a} 2x = \frac{x}{a}$.

$\left(\frac{\partial x}{\partial u}\right)_v$ - First express x as $f(u, v)$ then diff. w.r. to u keeping v constant.

- x is already given as a f^n of u & v.

$\therefore 2x \left(\frac{\partial x}{\partial u}\right)_v = a$

$\therefore \left(\frac{\partial x}{\partial u}\right)_v = \frac{a}{2x}$

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Now, $\left(\frac{\partial u}{\partial x}\right)_y \left(\frac{\partial x}{\partial u}\right)_v = \frac{x}{a} \cdot \frac{a}{2x} = \frac{1}{2}$ — (3)

Now, $\left(\frac{\partial v}{\partial y}\right)_x$ - First express v as $f(x,y)$ & then diff. w.r. to y keeping x as constant.

From (1) & (2),

$x^2 = au + bv$

$y^2 = au - bv$

$x^2 - y^2 = 2bv$

$\therefore v = \frac{x^2 - y^2}{2b}$

$\left(\frac{\partial v}{\partial y}\right)_x = \frac{1}{2b}(-2y) = -\frac{y}{b}$ — (4)

Now, $\left(\frac{\partial y}{\partial v}\right)_u$ - First express y as $f(u,v)$ then diff. w.r. to v keeping u constant.

$\therefore$ Here y is already given as f^n of u & v.

$\therefore 2y\left(\frac{\partial y}{\partial v}\right)_u = -b$

$\therefore \left(\frac{\partial y}{\partial v}\right)_u = \frac{-b}{2y}$ — (5)

From (4) & (5)

$\left(\frac{\partial v}{\partial y}\right)_x \left(\frac{\partial y}{\partial v}\right)_u = -\frac{y}{b} \cdot \frac{-b}{2y} = \frac{1}{2}$ — (6)

$\therefore$ From (3) & (6),

$\left(\frac{\partial u}{\partial x}\right)_y \left(\frac{\partial v}{\partial u}\right)_v = \frac{1}{2} = \left(\frac{\partial v}{\partial y}\right)_x \left(\frac{\partial y}{\partial v}\right)_u$.

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7) If $x = r\cos\theta$, $y = r\sin\theta$, prove that

i) $\left(\frac{\partial r}{\partial x}\right)_y = \left(\frac{\partial x}{\partial r}\right)_\theta$ ii) $r\left(\frac{\partial \theta}{\partial x}\right)_y = \frac{1}{r}\left(\frac{\partial x}{\partial \theta}\right)_r$.

i) Here $x = r\cos\theta$, $y = r\sin\theta$

$\therefore x^2 + y^2 = r^2$

$\left(\frac{\partial r}{\partial x}\right)_y$ - First express r as $f(x, y)$ then diff. w.r. to x keeping y constant

$\therefore 2x = 2r\left(\frac{\partial r}{\partial x}\right)_y$

$\therefore \left(\frac{\partial r}{\partial x}\right)_y = \frac{x}{r}$ — (1)

Again $x = r\cos\theta$

$\therefore x = f(r, \theta)$

$\left(\frac{\partial x}{\partial r}\right)_\theta = \cos\theta = \frac{x}{r}$ $(\because x = r\cos\theta)$ — (2)

$\therefore$ From (1) & (2),

$\left(\frac{\partial r}{\partial x}\right)_y = \left(\frac{\partial x}{\partial r}\right)_\theta$.

ii) $\left(\frac{\partial \theta}{\partial x}\right)_y$ - First express θ as $f(x, y)$ & diff w.r. to x keeping y constant.

Here $x = r\cos\theta$, $y = r\sin\theta$

$\therefore \frac{y}{x} = \tan\theta$.

$\therefore \theta = \tan^{-1} y/x$.

$\therefore \left(\frac{\partial \theta}{\partial x}\right)_y = \frac{1}{1 + \left(\frac{y}{x}\right)^2}\left(-\frac{y}{x^2}\right)$

$= \frac{x^2}{x^2 + y^2}\left(\frac{-y}{x^2}\right)$

$= -\frac{y}{x^2 + y^2} = \frac{-r\sin\theta}{r^2} = \frac{-\sin\theta}{r}$

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Now, $r \cdot \left(\frac{\partial \theta}{\partial x}\right)_y = r \cdot \left(\frac{-\sin\theta}{r}\right) = -\sin\theta$ — (3)

For $\left(\frac{\partial x}{\partial \theta}\right)_r$, $x = r\cos\theta$

$$\left(\frac{\partial x}{\partial \theta}\right)_r = -r\sin\theta$$

$$\therefore \frac{1}{r}\left(\frac{\partial x}{\partial \theta}\right)_r = -\sin\theta \quad \text{— (4)}$$

From (3) & (4),

$$r\left(\frac{\partial \theta}{\partial x}\right)_y = \frac{1}{r}\left(\frac{\partial x}{\partial \theta}\right)_r.$$

(3) If $ux + vy = 0$ and $\frac{u}{x} + \frac{v}{y} = 1$ then prove that $\left(\frac{\partial u}{\partial x}\right)_y - \left(\frac{\partial v}{\partial y}\right)_x = \frac{x^2+y^2}{y^2-x^2}$.

Here, $ux + vy = 0$ — (1)

$$\frac{u}{x} + \frac{v}{y} = 1 \quad \text{— (2)}$$

$\therefore$ From (1), $u = -\frac{vy}{x}$

$\therefore$ From (2), $\frac{-vy}{x^2} + \frac{v}{y} = 1$

$$\therefore v\left(\frac{1}{y} - \frac{y}{x^2}\right) = 1$$

$$\therefore v\left(\frac{x^2-y^2}{x^2y}\right) = 1$$

$$\therefore v = \frac{x^2y}{x^2-y^2}$$

Now, $v = f(x, y)$

$$\therefore \left(\frac{\partial v}{\partial y}\right)_x = \frac{(x^2-y^2)\cdot x^2 - x^2y(-2y)}{(x^2-y^2)^2}$$

$$= x^4 - x^2y^2 + 2x^2y^2 / (x^2-y^2)^2$$

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$$\left(\frac{\partial v}{\partial y}\right)_x = \frac{x^4 + x^2y^2}{(x^2-y^2)^2} \quad \text{— (3)}$$

Again, $u = -\frac{vy}{x}$

$$\therefore u = -\frac{y}{x}\left(\frac{x^2y}{x^2-y^2}\right)$$

$$\therefore u = -\frac{xy^2}{x^2-y^2}$$

$$\therefore u = \frac{xy^2}{y^2-x^2}$$

$$\therefore u = f(x, y)$$

$$\therefore \left(\frac{\partial u}{\partial x}\right)_y = \frac{y^2(y^2-x^2) - xy^2(-2x)}{(y^2-x^2)^2}$$

$$= \frac{y^4 - x^2y^2 + 2x^2y^2}{(y^2-x^2)^2}$$

$$= \frac{y^4 + x^2y^2}{(y^2-x^2)^2} \quad \text{— (4)}$$

From (3) & (4),

$$\left(\frac{\partial u}{\partial x}\right)_y - \left(\frac{\partial v}{\partial y}\right)_x = \frac{y^4 + x^2y^2}{(y^2-x^2)^2} - \frac{x^4 + x^2y^2}{(x^2-y^2)^2}$$

$$= \frac{y^4 - x^4 + x^2y^2 - x^2y^2}{(y^2-x^2)^2}$$

$$= \frac{(y^2-x^2)(y^2+x^2)}{(y^2-x^2)^2} = \frac{x^2+y^2}{y^2-x^2}.$$

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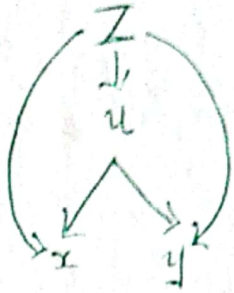
Total Derivatives:

Chain Rule:

If $z = f(u)$, where u is again a function of two variables x & y i.e. $u = \phi(x, y)$ then

$$\frac{\partial z}{\partial x} = \frac{dz}{du}\frac{\partial u}{\partial x} \quad \text{or} \quad \frac{df}{du}\frac{\partial u}{\partial x} \quad \text{or} \quad f'(u)\frac{\partial u}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{dz}{du}\cdot\frac{\partial u}{\partial y} \quad \text{or} \quad \frac{df}{du}\frac{\partial u}{\partial y} \quad \text{or} \quad f'(u)\frac{\partial u}{\partial y}.$$



Example:

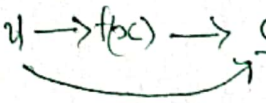
(1) For $z = \tan^{-1}\left(\frac{x}{y}\right)$, $x = u\sin v$, $y = u\cos v$, evaluate $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ at the point $(1.3, \frac{\pi}{6})$

Total Derivatives:

Chain Rule: One Independent variable & one intermediate variable.

When $u = f(x)$ is a differentiable fⁿ of x & $x = g(t)$ is a differentiable fⁿ of t, then u become a differentiable fⁿ of t and

$$\frac{du}{dt} = \frac{du}{dx}\cdot\frac{dx}{dt}.$$



chain rule for function of two Independent variables: 2 1 Intermediate variable

If $u = f(x, y)$, where $x = \phi(t)$, $y = \phi(t)$ then u is a fⁿ of t and is called the composite function of a single variable t.

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt}.$$



is called the total derivative of u.

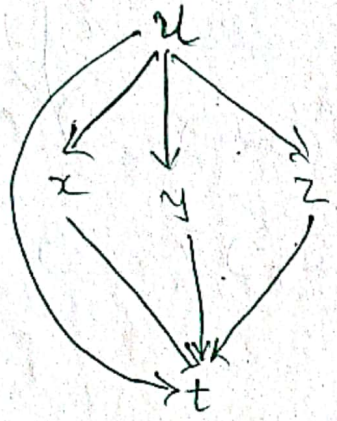
In differential form, $du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$ is called Total differential

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Chain rule for function of three independent variables:

If $u = f(x, y, z)$ and $x = \phi(t)$, $y = \psi(t)$, $z = \chi(t)$ then u is a function of t alone.

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}.$$



Examples:

① If $u = y^2 - 4ax$, $x = at^2$, $y = 2at$, Find $\frac{du}{dt}$.

Here, $u = y^2 - 4ax$, $x = at^2$, $y = 2at$



$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$$

$$= (-4a)(2at) + (2y)(2a)$$

$$= -8a^2t + 4a(2at)$$

$$= -8a^2t + 8a^2t$$

$$= 0.$$

② If $u = xy + yz + zx$ where $x = \frac{1}{t}$, $y = e^t$, $z = e^{-t}$, Find $\frac{du}{dt}$

Here, $u = xy + yz + zx$, $x = \frac{1}{t}$, $y = e^t$, $z = e^{-t}$

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}$$

$$= (y+z)\left(-\frac{1}{t^2}\right) + (x+z)(e^t) + (y+x)(-e^{-t})$$

$$= (e^t + e^{-t})\left(-\frac{1}{t^2}\right) + \left(\frac{1}{t} + e^{-t}\right)(e^t) + \left(e^t + \frac{1}{t}\right)e^{-t}$$

$$= -\frac{1}{t^2}(e^t + e^{-t}) + \frac{1}{t}(e^t - e^{-t})$$

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(3) If $z = x^2y + 3xy^4$ where $x = \sin 2t$, $y = \cos t$ find $\frac{dz}{dt}$ when $t=0$.

Here, $z = x^2y + 3xy^4$, $x = \sin 2t$, $y = \cos t$



$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$= (2x + 3y^4)(2\cos 2t) + (x^2 + 12xy^3)(-\sin t)$$

$$\left.\frac{dz}{dt}\right|_{t=0} = (2x + 3y^4)(2\cdot\cos 0) + (x^2 + 12xy^3)(-\sin 0)$$

$$= \cancel{(2x + 3y^4)}$$

At $t=0$, $x=0$, $y=1$

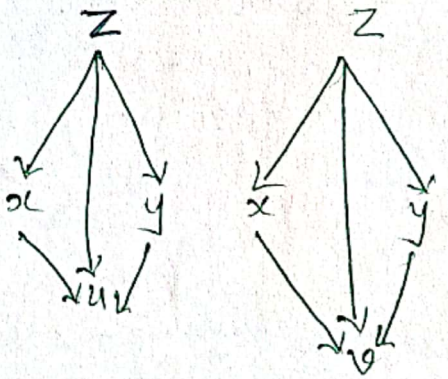
$$\left|\frac{dz}{dt}\right|_{t=0} = 2(0) + 3(1)^4(2) + 0 = 6.$$

Chain rule for function of two independent variables and two intermediate variables: (Composite f^n of two variables):

If $z = f(x,y)$ and $x = \phi(u,v)$, $y = \psi(u,v)$ then z is a function of u,v and z is called the composite function of two variables u and v.

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}.$$



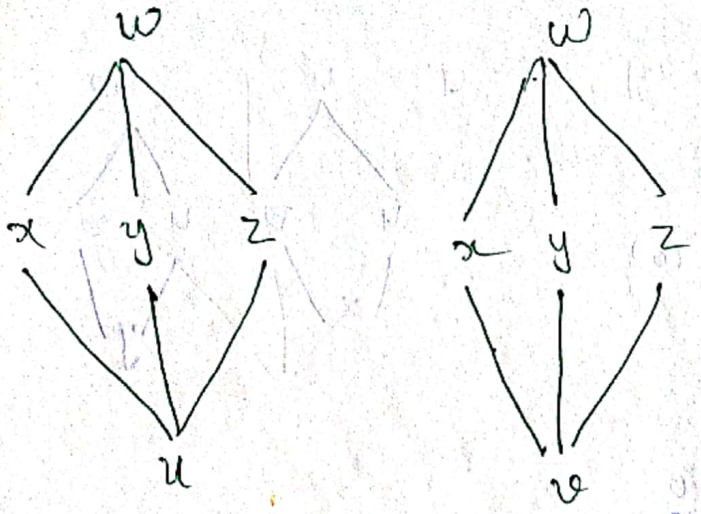
Chain rule for function of three independent variables and two intermediate variables: (Composite function of three variables):

If $w = f(x,y,z)$ and $x = \phi(u,v)$, $y = \psi(u,v)$, $z = \xi(u,v)$ then w is a function of u,v and w is called the composite function of two variables u,v.

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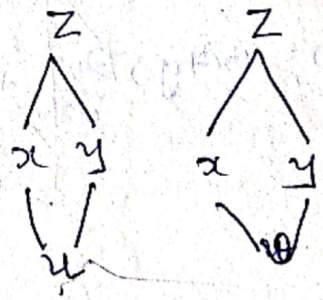
$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial u}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial v}.$$



Examples:

① If $z = f(x,y)$ and $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$, show that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$



$z = f(x,y)$, $x = e^u + e^{-v}$, $y = e^{-u} - e^v$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$

$$= \frac{\partial z}{\partial x}e^u + \frac{\partial z}{\partial y}(-e^{-u}) \quad \text{—— ①}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

$$= \frac{\partial z}{\partial x}(-e^{-v}) + \frac{\partial z}{\partial y}(-e^v) \quad \text{— ②}$$

From ① & ②

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}e^u - e^{-u}\frac{\partial z}{\partial y} + \frac{\partial z}{\partial x}e^{-v} + \frac{\partial z}{\partial y}e^v$$

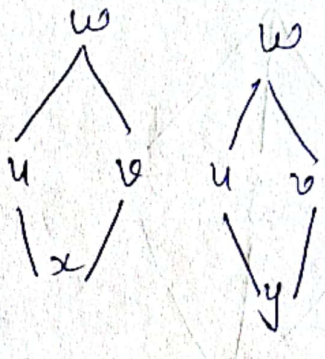
$$= \frac{\partial z}{\partial x}(e^u + e^{-v}) - \frac{\partial z}{\partial y}(e^{-u} - e^v)$$

$$= x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}.$$

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(2) If $w = \phi(u,v)$, $u = x^2 - y^2 - 2xy$, $v = y$, prove that $\frac{\partial w}{\partial v} = 0$ is equivalent to $(x+y)\frac{\partial w}{\partial x} + (x-y)\frac{\partial w}{\partial y} = 0$.

$w = \phi(u,v)$, $u = x^2 - y^2 - 2xy$, $v = y$



$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial x}$$

$$= \frac{\partial w}{\partial u}(2x - 2y) + \frac{\partial w}{\partial v}(0)$$

$$= \frac{\partial w}{\partial u}(2x - 2y)$$

Now, $$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial y}$$

$$= \frac{\partial w}{\partial u}(-2y - 2x) + \frac{\partial w}{\partial v}(1)$$

$$= -2(x+y)\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v}$$

Now, $(x+y)\frac{\partial w}{\partial x} + (x-y)\frac{\partial w}{\partial y} = 2(x+y)(x-y)\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v} - 2(x-y)(x+y)\frac{\partial w}{\partial u}$

$= \frac{\partial w}{\partial v}$

$= (x+y)\left\{\frac{\partial w}{\partial u}2(x-y)\right\} + (x-y)\left\{-2(x+y)\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v}\right\}$

$= 2(x+y)(x-y)\frac{\partial w}{\partial u} - 2(x+y)(x-y)\frac{\partial w}{\partial u} + (x-y)\frac{\partial w}{\partial v}$

$= (x-y)\frac{\partial w}{\partial v}$.

If $\frac{\partial w}{\partial v} = 0$ then $(x+y)\frac{\partial w}{\partial x} + (x-y)\frac{\partial w}{\partial y} = 0$

Hence, $\frac{\partial w}{\partial v} = 0$ is equivalent to $(x+y)\frac{\partial w}{\partial x} + (x-y)\frac{\partial w}{\partial y} = 0$

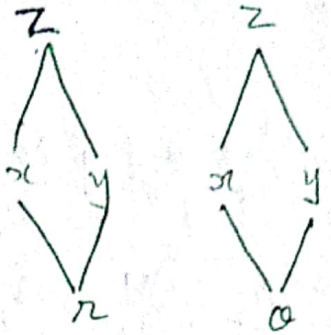
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③ If $x = r\cosh\theta$, $y = r\sinh\theta$, show that $(z_x - z_y)(x-y) = r z_r - z_\theta$.

$z = f(x, y)$, $x = r\cosh\theta$, $y = r\sinh\theta$.

$$z_r = \frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r}$$

$$= \frac{\partial z}{\partial x}\cosh\theta + \frac{\partial z}{\partial y}\sinh\theta$$



$$z_\theta = \frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial \theta}$$

$$= \frac{\partial z}{\partial x} r\sinh\theta + \frac{\partial z}{\partial y} r\cosh\theta$$

Now, $r \cdot z_r - z_\theta = \frac{\partial z}{\partial x} r\cosh\theta + \frac{\partial z}{\partial y} r\sinh\theta - \frac{\partial z}{\partial x} r\sinh\theta - \frac{\partial z}{\partial y} r\cosh\theta$

$$= \frac{\partial z}{\partial x}x + \frac{\partial z}{\partial y}y - \frac{\partial z}{\partial x}y - \frac{\partial z}{\partial y}x$$

$$= \frac{\partial z}{\partial x}(x-y) - \frac{\partial z}{\partial y}(x-y)$$

$$= (x-y)\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right).$$

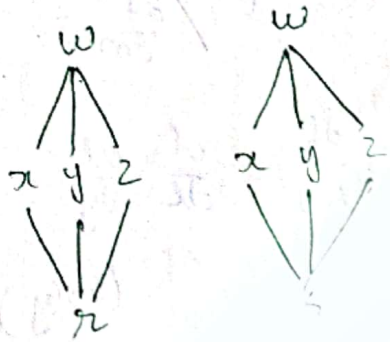
$$= (x-y)(z_x - z_y).$$

④ Find $\frac{\partial w}{\partial r}, \frac{\partial w}{\partial s}$ in terms of r and s if $w = x + 2y + z^2$, where $x = \frac{r}{s}$, $y = r^2 + \log s$, $z = 2r$.

$x = \frac{r}{s}$, $y = r^2 + \log s$, $z = 2r$

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r} \quad \text{--- (1)}$$

$\frac{\partial x}{\partial r} = \frac{1}{s}$, $\frac{\partial y}{\partial r} = 2r$, $\frac{\partial z}{\partial r} = 2$



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∴ From ①, $\frac{\partial w}{\partial r} = 1\cdot\left(\frac{1}{s}\right) + 2(2r) + (2z)\cdot(2)$

$= \frac{1}{s} + 4r + 4z$

$= \frac{1}{s} + 4r + 4(2r)$

$= \frac{1}{s} + 12r$

$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s}$ — ②

$\frac{\partial x}{\partial s} = -\frac{r}{s^2}$, $\frac{\partial y}{\partial s} = \frac{1}{s}$, $\frac{\partial z}{\partial s} = 0$

∴ From ②, $\frac{\partial w}{\partial s} = (1)\left(-\frac{r}{s^2}\right) + 2\left(\frac{1}{s}\right) + 2z(0)$

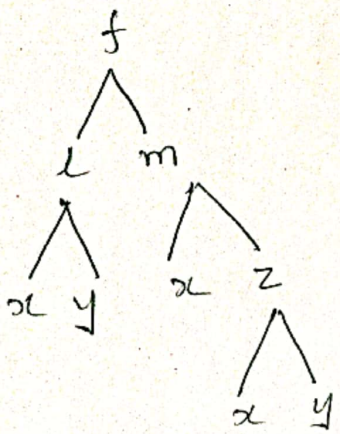
$= -\frac{r}{s^2} + \frac{2}{s}$

Before 5th example, start example ④.

⑤ If $f(xy^2, z-2x) = 0$, show that $2x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y} = 4x$.

Let, $l = xy^2$, $m = z-2x$

∴ $f(l, m) = 0$



$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial l}\frac{\partial l}{\partial x} + \frac{\partial f}{\partial m}\frac{\partial m}{\partial x}$

$0 = \frac{\partial f}{\partial l}(y^2) + \frac{\partial f}{\partial m}\left(-2 + \frac{\partial z}{\partial x}\right)$

∴ $\frac{\partial f}{\partial l}y^2 = -\frac{\partial f}{\partial m}\left(-2 + \frac{\partial z}{\partial x}\right)$

∴ $\frac{\partial f}{\partial l} \Big/ \frac{\partial f}{\partial m} = \frac{2 - \frac{\partial z}{\partial x}}{y^2}$ — ①

and $\frac{\partial f}{\partial y} = \frac{\partial f}{\partial l}\frac{\partial l}{\partial y} + \frac{\partial f}{\partial m}\frac{\partial m}{\partial y}$

$0 = \frac{\partial f}{\partial l}(2xy) + \frac{\partial f}{\partial m}\left(\frac{\partial z}{\partial y}\right)$

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$$\therefore \frac{\partial f/\partial x}{\partial f/\partial m} = -\frac{\partial z/\partial y}{2xy} \quad \text{— (2)}$$

From (1) & (2),

$$\frac{2-\frac{\partial z}{\partial x}}{y^2} = -\frac{\partial z/\partial y}{2xy}$$

$$\therefore 2-\frac{\partial z}{\partial x} = -\frac{\partial z}{\partial y}\frac{y}{2z}$$

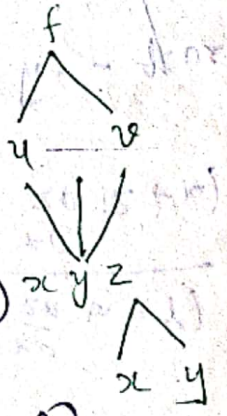
$$\therefore 4x - 2x\frac{\partial z}{\partial x} = -y\frac{\partial z}{\partial y}$$

$$\therefore 2x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y} = 4x.$$

(6) $f(lx+my+nz, x^2+y^2+z^2) = 0$, prove that $(ly-mx)+(my-nz)\frac{\partial z}{\partial x}+(lz-nx)\frac{\partial z}{\partial y} = 0$.

Let, $lx+my+nz = u$, $v = x^2+y^2+z^2$

$\therefore f(u,v) = 0$



Now, $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x}$

$\therefore 0 = \frac{\partial f}{\partial u}\left(l + n\frac{\partial z}{\partial x}\right) + \frac{\partial f}{\partial v}\left(2x + 2z\frac{\partial z}{\partial x}\right)$

$\therefore \frac{\partial f}{\partial u}\left(l + n\frac{\partial z}{\partial x}\right) = -\frac{\partial f}{\partial v}\left(2x + 2z\frac{\partial z}{\partial x}\right)$ — (1)

And $\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y}$.

$0 = \frac{\partial f}{\partial u}\left(m + n\frac{\partial z}{\partial y}\right) + \frac{\partial f}{\partial v}\left(2y + 2z\frac{\partial z}{\partial y}\right)$

$\frac{\partial f}{\partial u}\left(m + n\frac{\partial z}{\partial y}\right) = -\frac{\partial f}{\partial v}\left(2y + 2z\frac{\partial z}{\partial y}\right)$ — (2)

$\therefore$ From (1) & (2),

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$$\frac{\partial f}{\partial u} \Big/ \frac{\partial f}{\partial v} = \frac{-2\left(x + z\frac{\partial z}{\partial x}\right)}{\left(l + n\frac{\partial z}{\partial x}\right)}$$

$$\partial f/\partial u \Big/ \frac{\partial f}{\partial v} = \frac{-2\left(y + z\frac{\partial z}{\partial y}\right)}{\left(m + n\frac{\partial z}{\partial y}\right)}$$

$$\therefore \frac{\left(x + z\frac{\partial z}{\partial x}\right)}{\left(l + n\frac{\partial z}{\partial x}\right)} = \frac{\left(y + z\frac{\partial z}{\partial y}\right)}{\left(m + n\frac{\partial z}{\partial y}\right)}$$

$$\therefore \left(x + z\frac{\partial z}{\partial x}\right)\left(m + n\frac{\partial z}{\partial y}\right) = \left(y + z\frac{\partial z}{\partial y}\right)\left(l + n\frac{\partial z}{\partial x}\right)$$

$$\therefore mx + nx\frac{\partial z}{\partial y} + mz\frac{\partial z}{\partial x} + nz\frac{\partial z}{\partial x}\frac{\partial z}{\partial y} = ly + ny\frac{\partial z}{\partial x} + lz\frac{\partial z}{\partial y} + nz\frac{\partial z}{\partial y}\frac{\partial z}{\partial x}$$

$$\therefore mx - ly + nx\frac{\partial z}{\partial y} - lz\frac{\partial z}{\partial y} + mz\frac{\partial z}{\partial x} - ny\frac{\partial z}{\partial x} = 0$$

$$\therefore \frac{\left(x + z\frac{\partial z}{\partial x}\right)}{\left(l + n\frac{\partial z}{\partial x}\right)} = \frac{\left(y + z\frac{\partial z}{\partial y}\right)}{\left(m + n\frac{\partial z}{\partial y}\right)}$$

$$\therefore \left(x + z\frac{\partial z}{\partial x}\right)\left(m + n\frac{\partial z}{\partial y}\right) = \left(y + z\frac{\partial z}{\partial y}\right)\left(l + n\frac{\partial z}{\partial x}\right)$$

$$\therefore mx + nx\frac{\partial z}{\partial y} + mz\frac{\partial z}{\partial x} + nz\frac{\partial z}{\partial x}\frac{\partial z}{\partial y} = ly + ny\frac{\partial z}{\partial x} + lz\frac{\partial z}{\partial y} + nz\frac{\partial z}{\partial y}\frac{\partial z}{\partial x}$$

$$\therefore mx - ly + nx\frac{\partial z}{\partial y} - lz\frac{\partial z}{\partial y} + mz\frac{\partial z}{\partial x} - ny\frac{\partial z}{\partial x} = 0$$

$$\therefore (ly - mx) + (ny - mz)\frac{\partial z}{\partial x} + (lz - nx)\frac{\partial z}{\partial y} = 0$$

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